

QUAKERTOWN, PA, USA

WMO: 720324

Lat: 40.435N

Lon: 75.382W

Elev: 526

StdP: 14.42

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 64753

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.1	11.7	-5.8	4.1	9.9	-0.3	5.5	15.9	18.6	30.2	16.8	31.4	3.9	310	0.428

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	90.7	74.6	88.4	73.3	85.8	72.1	77.4	86.4	75.8	83.5	74.4	81.5	5.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.9	133.3	80.6	73.2	125.7	78.7	72.4	122.5	77.9	41.2	85.1	39.5	83.1	38.3	81.8	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.2	13.4	11.8	1.8	95.8	5.7	3.4	-2.3	98.3	-5.6	100.3	-8.8	102.2	-13.0	104.7		
			-0.3	80.0	5.6	1.2	-4.3	80.9	-7.6	81.6	-10.7	82.2	-14.8	83.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.0	29.4	30.1	39.0	50.9	60.6	69.2	74.3	72.1	65.3	54.2	42.4	34.5
	DBStd	17.68	9.80	9.26	9.28	8.49	7.92	5.96	4.94	5.15	6.72	8.32	8.04	8.77
	HDD50	2444	638	560	360	88	7	0	0	0	0	51	252	488
	HDD65	5590	1102	978	807	430	186	28	2	6	78	350	677	946
	CDD50	3161	1	2	18	116	337	576	755	684	461	180	24	7
	CDD65	831	0	0	0	7	51	154	292	225	88	14	0	0
	CDH74	7661	0	0	6	133	571	1417	2744	1946	754	90	0	0
	CDH80	2591	0	0	0	30	166	439	1064	652	223	17	0	0
Wind	WSAvg	3.9	4.1	4.7	5.5	5.4	3.9	3.3	2.7	2.3	2.7	3.6	4.0	4.0
Precipitation	PrecAvg	48.1	3.6	2.6	4.2	3.8	3.9	4.2	4.9	4.2	4.6	4.5	3.3	4.3
	PrecMax	63.1	6.5	6.4	6.7	8.9	6.8	10.4	10.5	8.7	10.9	10.8	5.7	9.8
	PrecMin	38.6	1.5	0.9	0.9	1.2	1.3	1.4	0.7	1.5	0.4	0.2	1.2	1.2
	PrecStd	6.1	1.5	1.2	1.5	1.8	1.5	2.1	2.0	1.7	2.6	2.7	1.1	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.2	64.0	73.1	83.9	89.6	92.5	94.9	93.0	90.7	81.7	69.7	64.2
		MCWB	55.2	55.5	59.0	62.6	71.4	75.2	75.9	75.4	74.1	68.9	61.2	60.9
	2%	DB	54.6	55.5	65.6	77.3	84.3	88.3	91.0	89.9	85.9	75.4	64.4	57.1
		MCWB	50.8	47.5	55.3	60.6	69.7	73.2	74.8	74.9	71.5	66.2	57.4	54.2
	5%	DB	47.8	48.2	58.9	72.5	81.1	84.2	89.7	86.1	81.8	72.4	61.1	53.6
		MCWB	44.2	42.1	49.6	58.2	67.4	71.5	74.0	72.9	69.1	64.5	53.7	51.1
	10%	DB	44.7	43.3	54.4	66.5	75.5	81.8	85.9	83.6	78.7	68.3	56.6	47.8
		MCWB	41.0	38.2	47.3	55.1	64.3	69.1	72.5	71.4	67.4	62.0	51.1	44.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.1	58.9	61.9	67.4	74.9	77.1	79.7	79.0	77.2	73.1	63.3	61.6
		MCDB	59.4	63.8	69.8	73.6	83.4	87.8	89.9	89.4	85.2	78.3	66.7	63.6
	2%	WB	51.7	49.6	56.6	63.7	72.0	75.1	77.7	77.2	74.4	70.2	59.7	55.4
		MCDB	54.1	57.3	63.2	72.9	81.2	83.9	86.9	85.8	80.4	73.4	63.6	56.9
	5%	WB	45.4	42.7	52.0	60.5	69.7	73.4	76.0	75.3	72.5	66.9	55.6	50.3
		MCDB	46.7	47.3	57.2	69.6	78.2	81.6	84.1	81.8	78.1	71.2	59.2	51.9
	10%	WB	40.6	38.4	47.4	57.2	67.2	71.9	74.6	73.5	70.5	63.2	52.3	45.2
		MCDB	43.7	42.8	53.3	65.2	73.5	79.1	82.0	79.5	75.9	65.8	55.8	46.7
Mean Daily Temperature Range	5% DB	MDBR	15.4	16.8	18.5	21.9	21.7	20.4	20.4	20.3	21.0	19.5	18.4	14.8
		MCDBR	22.9	25.1	27.1	30.2	27.1	24.2	23.9	22.9	25.1	24.2	24.6	20.3
		MCWBR	20.5	18.8	18.5	15.9	13.5	10.8	9.1	9.6	12.1	14.1	17.7	18.7
	5% WB	MCDBR	21.0	23.3	23.9	24.4	23.7	20.8	19.3	18.7	19.9	17.3	19.8	18.6
		MCWBR	19.8	18.9	19.5	15.8	12.9	10.7	8.6	8.8	10.7	12.4	16.8	18.8
Clear-Sky Solar Irradiance	taub	0.303	0.319	0.355	0.385	0.428	0.487	0.486	0.481	0.396	0.356	0.322	0.303	
	taud	2.453	2.390	2.369	2.312	2.221	2.096	2.113	2.113	2.322	2.447	2.505	2.504	
	Ebn at Noon	273	284	284	282	270	254	253	250	267	267	264	265	
	Edh at Noon	25	30	35	40	44	50	49	48	36	29	24	22	
All-Sky Solar Radiation	RadAvg	582	870	1150	1496	1669	1821	1875	1620	1344	944	663	497	
	RadStd	51	97	89	142	196	153	94	104	167	102	60	54	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (68 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	+216	+122

Nomenclature: See separate page